Program Description

Since the 1990s, Colorado School of Mines has been a leading institution for the study of space resources and their utilization through its *Center for Space Resources*. It has also become a destination for space scientists and engineers, government agencies, aerospace companies, entrepreneurs, the mining and minerals industry, financial and legal experts, and policy makers to discuss all topics related to this field at its annual *Space Resources Roundtable*.

In recent years, growing interest in the identification, extraction, and utilization of space resources by space agencies and the private sector has been driven by an awareness that further exploration and growth of commercial opportunities in space will require extraction of materials, production of propellants, and power generation from extraterrestrial resources for more affordable and flexible transportation, facilities construction, energy production, manufacturing of parts, and life support. The broad topic of space resources brings together many fields in which Mines has a strong presence, including remote sensing, geomechanics, mining, materials/metallurgy, robotics/automation, advanced manufacturing and construction, electrochemistry, solar and nuclear energy, resource economics, and public policy.

In this light, Mines launched a first-of-its-kind multi-disciplinary graduate program in Space Resources in 2017 to offer Post-Baccalaureate Certificate, Master of Science, and Ph.D. degrees for college graduates and professionals interested in this emerging arena. The program focuses on developing core knowledge and gaining design practices in systems for responsible exploration, extraction, use, and stewardship of resources in the Solar System.

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For more information on the Space Resources program, visit https://space.mines.edu.

Space Resources Program Requirements

The interdisciplinary Space Resources program is targeted to train recent graduates, as well as professionals interested in expanding their knowledge and skills to address the opportunities and challenges in space resource exploration, extraction, and utilization. Space Resources touches on physical sciences, engineering, and the social science fields. Thus, this program engages faculty members from many academic departments at Mines. Graduates from the Space Resources program will be prepared to serve the growing interest of industry, government, and academia to identify, extract, process, and utilize resources beyond Earth. Students will have a broad, multi-disciplinary understanding of the overall flow of activities in the development of extraterrestrial resources, a high-level exposure to the different science, engineering, economics, and business disciplines involved in each phase, and an understanding of the current status in the space resources field across academia, government, and the private sector.

The graduate program for Space Resources includes the following degree options:

- a 12-credit-hour graduate certificate offered online,
- a 31-credit Master of Science Non-Thesis (MS-NT) degree in Space Resources offered online,
- a PhD program in Space Resources requiring a total of 72 credits of combined coursework and research, and a doctoral dissertation. The PhD program can be completed on campus and also online for those students approved to conduct their research remotely by their advisor and dissertation committee.

Mines' Combined Undergraduate/Graduate Degree Program

With approval of the Space Resources Program faculty, students may be allowed to enroll in a combined undergraduate/graduate program in Space Resources. Students approved to enroll in this program may double count up to six credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with "B" or better, not be substitutes for required coursework, and meet all other University, Department, and Program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

Graduate Certificate

The Graduate Certificate is intended to give the student a panoramic view of space resources to understand the role that scientific knowledge, engineering systems, economic analyses, business cases, and legal and policy aspects play in this field. This option requires students to take a minimum of 12 credits online with 3 credits from Space Resources Fundamentals (SPRS501) and 9 credits chosen from other courses. Table 1 lists the courses that will comprise the curriculum for the Graduate Certificate.

Table 1 – Required courses for 12-credit hour Graduate Certificate in Space Resources

SPRS501	SPACE RESOURCES FUNDAMENTALS	3.0
ELECTIVES	SPRS 502 or Elective courses chosen from Table 3	9.0
Total Semester Hrs		**12.0**

Elective courses can be taken from a variety of important topics on space resources, such as: planetary geology, space operations, remote sensing, resource economics, materials extraction, advanced manufacturing, space law and policy, and other topics which continue to be introduced as relevant subjects are identified and new courses developed (see Table 3). Students are also allowed to take courses from other departments and programs with the approval from the Space Resources Program Director in consultation with faculty members of the program.

Master of Science (Non-Thesis)

The Master of Science program is intended to give the degree holder the knowledge and skills to make immediate contributions to any government agency or company pursuing technical activities related to space resources. The Master of Science degree program is exclusively non-thesis (MS-NT) and online. The MS-NT degree program coursework requires 31 credits as laid out in Table 2, with 13 credits from five courses (SPRS501, SPRS502, SPRS503, SPRS591 and SPRS592) and 18 credits from elective courses. For students coming into the program with previous Master or PhD degrees, up to 9 credits of relevant courses can be transferred after approval from the Space Resources program director in consultation with faculty members of the program.

Table 2 – Required courses for the 31-credit hour online Master of Science Non-Thesis degree in Space Resources

SPRS501	SPACE RESOURCES FUNDAMENTALS	3.0
SPRS502	SPACE SYSTEMS ENGINEERING	3.0
SPRS503	SPACE RESOURCES SEMINAR	1.0
SPRS591	SPACE RESOURCES PROJECT I	3.0
SPRS592	SPACE RESOURCES PROJECT II	3.0
ELECTIVES	Elective courses chosen from Table 3	18.0
Total Semester Hrs		**31.0**

The 6 credits of project courses (SPRS 591 and 592) allow students to develop a design, system, or economic analysis focused on the exploration, extraction, utilization, and responsible stewardship of space resources. The 18 credits of elective courses can be selected from any of the various courses listed in Table 3. These courses provide students with a broad interdisciplinary coverage of critical areas in space resources, such as: Remote Sensing and Resource Assessment; Planetary Geology; Extraction, Processing, and Resource Utilization; Power and Energy; Robotics, Autonomy, and Communications; and Economics, Law, and Policy.

Additionally, new courses continue to be developed and introduced on various relevant subjects. Also, courses from other departments and programs at Mines are allowed as electives after approval from the Space Resources program director in consultation with faculty members of the program.

Table 3 – Current elective courses for all degree options of the Space Resources program (additional topics continue to be introduced as relevant subjects are identified and new courses developed). Other courses at Mines are also allowed after approval from the Space Resources program director in consultation with faculty members of the program.

AMFG501	ADDITIVE MANUFACTURING PROCESSES	3.0
AMFG521	DESIGN FOR ADDITIVE MANUFACTURING	3.0
AMFG522	LEAN MANUFACTURING	3.0
CEEN501	LIFE CYCLE ASSESSMENT	3.0
GEOL557	EARTH RESOURCE DATA SCIENCE 1: FUNDAMENTALS	3.0
GEOL558	EARTH RESOURCE DATA SCIENCE 2: APPLICATIONS AND MACHINE-LEARNING	3.0
MEGN551	ADVANCED FLUID MECHANICS	3.0
MEGN566	COMBUSTION	3.0
MEGN571	ADVANCED HEAT TRANSFER	3.0
SPRS504	ECONOMICS OF SPACE RESOURCES	3.0
SPRS505	SPACE OPERATIONS	3.0
SPRS506	INTERNATIONAL SPACE LAW & POLICY	3.0
SPRS507	ADVANCED PLANETARY GEOLOGY	3.0
SPRS508	REGOLITH PROPERTIES AND PROCESSING	3.0
SPRS509	SPACE ROBOTICS	
NUGN598	SPECIAL TOPICS (Introduction to Space Nuclear Technology)	3.0
SPRS598B	INNOV8X – ENTREPRENEURSHIP AND INNOVATION	3.0
SPRS599	INDEPENDENT STUDY IN SPACE RESOURCES	1-6

PhD in Space Resources

The Doctor of Philosophy (PhD) is intended to provide the degree holders the necessary knowledge and research skills to pursue academic positions or contribute as a space-resources specialist in industry or government agencies. A student who completes a PhD in Space Resources will possess all the training of a Master's degree holder with further specialization in one or more areas within the space resources field. The completed doctoral dissertation will make original contributions to the field.

1) General Requirements

The Doctor of Philosophy (PhD) degree in Space Resources requires 72 total credits, consisting of at least 36 credits of courses beyond the BS and at least 24 research credits. The balance of 12 credits can be either course or research credits. Coursework beyond the MS degree program will not be restricted other than approved by the student's advisor and Dissertation Committee. The PhD degree allows for both on-campus and online options. The latter requires approval by the student advisor, Dissertation Committee, and Program Director who will review and approve if the research project can be conducted remotely. If a student requests to transfer from on-campus to online status (or vice versa) after passing the PhD Qualifying Exam, the Dissertation Committee will also have to approve the change in addition to the advisor.

For students coming into the PhD program with a Master Non-Thesis degree in an engineering, science, economics, or other relevant field, up to 24 credits of courses can be transferred, after approval from the Space Resources Program Director in consultation with the program faculty. These students will still be required to take 13 credits from five courses (SPRS501, SPRS502, SPRS503, SPRS591 and SPRS592) and other elective courses needed to complete the minimum 36 credit course requirement. For students with a previous Master with Thesis or PhD degree, up to 36 credits of relevant courses can be approved for transfer; however, the student will still be required to take the 13 credits from the previously mentioned five courses and other elective courses needed to complete the minimum 36 credit course requirement. For all students starting the PhD program already with a Mines Space Resources Certificate or MS-NT degree (or transferring to the PhD from any of these two programs), all credits taken at the Mines Space Resources program can be transferred to the PhD degree.

Students in the Space Resources PhD degree program must successfully complete a Qualifying Exam, defend a written Dissertation Research Proposal, submit at least one publication to a peer-reviewed journal, deliver conference and program seminar presentations, and write and defend a Doctoral Dissertation, in addition to their course and research work. PhD research is aimed at advancing fundamental knowledge in the Space Resources field. From the beginning of their studies at Mines, students must have an advisor to direct and monitor their academic plan, research, and independent studies. A PhD Dissertation Committee must also be assembled after passing the Qualifying Exam, as outlined below.

2) PhD Qualifying Exam

Students enrolled in the Space Resources PhD program are required to pass a Qualifying Exam. The PhD Qualifying Exam will be administered at specific dates during the year as determined by the Space Resources PhD Qualifying Exam Committee. The Committee will appoint an Exam Chair, who oversees the process and ensures that the Exam is administered fairly. The Qualifying Exam assesses attributes expected of a successful PhD student, including: